CHAPTER 1:

Introduction

The global aquaculture sector has emerged as a critical driver of food security and economic development, accounting for approximately half of the world's seafood production (Apine et al., 2023). Within this rapidly expanding industry, mud crab (Scylla species) farming represents a particularly lucrative yet challenging agricultural enterprise, especially in tropical coastal regions of Southeast Asia, India, and the Pacific Islands. However, mud crab aquaculture faces significant operational and environmental obstacles that severely limit productivity and sustainability. The primary challenges include cannibalistic behavior among crabs, sudden water quality fluctuations caused by environmental shifts, and the lack of real-time monitoring systems to detect and prevent mortality events (Apine et al., 2023).

Traditional mud crab farming relies heavily on manual monitoring and reactive management approaches. Farmers discover problems only after significant losses have occurred, often when water quality has deteriorated or cannibalism has decimated populations. Recent research indicates that high mortality rates and limited seed supply remain the most critical barriers to achieving sustainable mud crab production (Apine et al., 2023). Additionally, disease outbreaks, particularly the Mud Crab Reovirus (MCRV) and bacterial infections, cause chronic mass mortalities, with survival rates often dropping below critical thresholds in open culture systems (Nivas et al., 2023). The economic vulnerability of small-scale farmers is further compounded by the lack of access to governmental support and financing, which prevents them from adopting improved technologies or implementing better management practices.

The emergence of Internet of Things (IoT) technology and smart aquaculture systems offers transformative potential for addressing these challenges. Recent studies demonstrate that IoT-based monitoring systems can significantly enhance aquaculture productivity by enabling real-time tracking of critical water quality parameters such as

dissolved oxygen, pH, temperature, ammonia levels, and salinity (Shete et al., 2025; Olanubi et al., 2024). Furthermore, sensor-based automated feeding systems have been shown to reduce feed waste by up to 25% while improving growth rates (AKVA Group, 2023). However, most existing IoT solutions have been developed for large-scale, industrialized aquaculture operations and remain prohibitively expensive and complex for smallholder farmers in developing countries who lack reliable grid electricity and internet connectivity (Roy et al., 2024). The global IoT aquaculture market is projected to expand from USD 923.59 million in 2025 to USD 1.72 billion by 2026, yet accessibility remains a significant barrier for rural coastal communities (Global Growth Insights, 2025).

LoRa (Long Range) is a low-power wireless radio technology that uses Chirp Spread Spectrum signals to transmit data over long distances while resisting electrical noise and interference. The Semtech SX1262 transceiver chosen for this project provides high receiver sensitivity down to −148 dBm and a transmit power of up to +22 dBm, giving it a strong signal budget for remote communication (Semtech Corporation, 2024). While these specifications represent ideal chip ratings, the real-world range depends on radio settings, antenna quality, and local transmission limits set by the National Telecommunications Commission. In practical testing, Pires and Gomes (2024) demonstrated that a direct point-to-point LoRa link could stream water quality readings over 2.53 km without requiring a central network server. Similarly, Syed Taha et al. (2024) showed that fine-tuning radio parameters and antennas lowered data loss from 22% down to 5.4% across a 1,560-meter water channel. Because this setup only needs to transfer telemetry across a single link between the pond and the farmhouse, a direct point-to-point LoRa connection was adopted to eliminate the hardware cost, complex setup, and service subscriptions associated with multi-channel LoRaWAN gateways.

This study presents the Solar Automated LoRa-Integrated Mud Crab Aeration Network with Crab Grid Observation (S.A.L.I.M.A.N.G.O.), a prototype built for mud crab (*Scylla serrata*) fattening in off-grid mangrove areas. The system isolates eight crabs in

individual slotted boxes on a floating platform to prevent cannibalism. A submerged, centrally located sensor hub continuously measures dissolved oxygen, pH, temperature, and salinity. Aeration runs continuously by default and is automatically forced on whenever dissolved oxygen falls below a critical safety threshold, protecting the stock during dangerous overnight oxygen crashes. Instead of placing delicate solar panels or batteries on the water, the floating platform draws power from a protected 220 V AC line connected to the home's existing solar array, which is stepped down to 12 V DC on the float. The float's microcontroller executes all emergency decisions locally and sends telemetry and a 60-second status heartbeat over a 915 MHz LoRa radio to a receiver at the house, which also forwards commands sent from an online dashboard. By combining individual housing, continuous sensing, and automatic life-support aeration, S.A.L.I.M.A.N.G.O. transforms mud crab fattening from manual, reactive guessing into automated, data-driven management.

Theoretical Framework

This study addresses the high mortality rates in mud crab aquaculture by analyzing both environmental fluctuations and biological vulnerabilities. It is anchored on two primary theories that justify the system's multimodal approach:

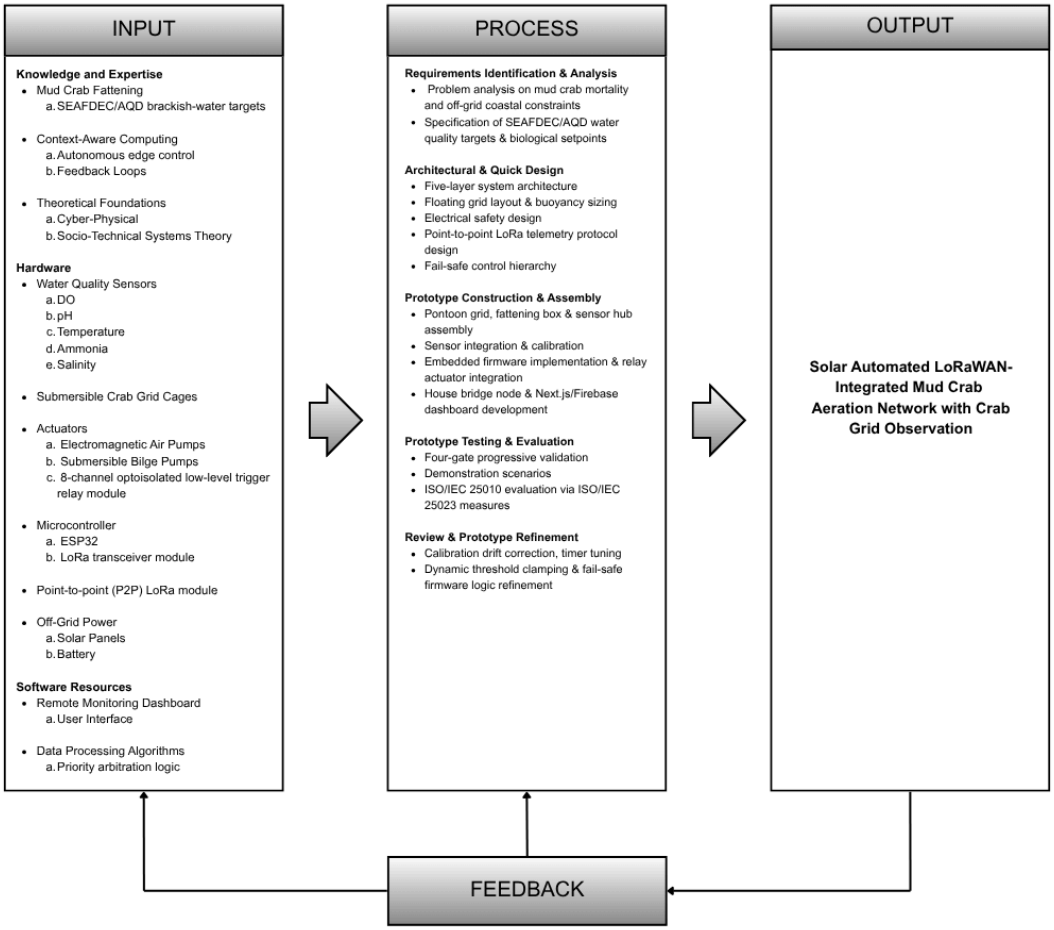
Cyber-Physical Systems Theory

This theory states that complex agricultural environments can be optimized when physical and biological processes are seamlessly integrated with computational algorithms and automated networks. In traditional setups, environmental shifts like heavy rainfall cause sudden drops in salinity, while unseen spikes in ammonia or drops in dissolved oxygen (DO) lead to toxic conditions and suffocation. By utilizing a multiparameter sensor array to track these hidden chemic/1al variables, the system acts as an artificial environmental regulator. When hazardous parameters are detected, the embedded logic triggers automated physical actuators, such as a dynamic pump that provides water circulation and can support a controlled salinity experiment or solar-powered aerators to restore oxygen

circulation. This allows the system to physically correct the ecosystem before a mass mortality event occurs.

Socio-Technical Systems Theory

This theory posits that a technological solution can only succeed if its infrastructure is reciprocally optimized for the social and economic constraints of its target users. Small-scale aquaculture operators often farm in remote coastal or mangrove areas where standard internet is unavailable and modern indoor alternatives are prohibitively expensive and require high energy consumption. By housing each crab in an individual slotted box on a floating platform to prevent cannibalism, and by pairing this design with point-to-point LoRa connectivity, the system achieves long-range telemetry with low power consumption and without pond-side internet. This design choice bridges the gap between active management and the off-grid realities of rural farmers, supporting the shift of mud crab fattening from a reactive, labor-intensive practice toward a proactive, data-driven one that protects farmer investments.



Conceptual Framework

Figure x: Conceptual Framework of the Study

The conceptual framework of this study follows the Input–Process–Output (IPO) model. The model shows how mud crab fattening knowledge and off-grid Internet of Things (IoT) resources are combined to develop a context-aware, automated system. The system aims to prevent cannibalism, reduce crab mortality caused by poor water quality, and support rural coastal farmers who work without grid electricity or internet access at the pond.

The input consists of knowledge and expertise, hardware, and software resources. The knowledge inputs include mud crab fattening practices based on the brackish-water targets of SEAFDEC/AQD, context-aware computing through autonomous edge control and feedback loops, and two theoretical foundations: Cyber-Physical Systems theory and Socio-Technical Systems theory. The hardware includes water quality sensors for dissolved oxygen, pH, temperature, and salinity; eight floating crab fattening boxes that keep each crab separate; air pumps for aeration and bilge pumps for water circulation, both controlled through a relay module; ESP32 microcontrollers paired with LoRa transceiver modules for point-to-point communication between the pond and the house; and a solar power source, provided during testing by the household's existing solar installation. The software resources include a remote monitoring dashboard and data processing algorithms that decide which control action takes priority and estimate ammonia toxicity risk from pH and temperature readings.

The process follows the prototyping model, which has five stages. In the Requirements Identification and Analysis stage, the researchers study the causes of mud crab mortality and the limitations of off-grid coastal sites, and set the water quality targets the system must maintain. In the Architectural and Quick Design stage, the researchers plan the system architecture, the layout and buoyancy of the floating grid, the electrical safety measures, the point-to-point LoRa communication, and a fail-safe control order in which low dissolved oxygen always triggers aeration. In the Prototype Construction and Assembly stage, the researchers build the floating grid and sensor hub, calibrate the sensors, program the controllers, and develop the web dashboard. In the Prototype Testing and Evaluation stage, the researchers test the prototype in four progressive stages, from bench testing to a stocked pond, run demonstration scenarios, and evaluate the system using the ISO/IEC 25010:2011 characteristics of Functional Suitability, Performance Efficiency, Usability, Reliability, and Maintainability, measured through ISO/IEC 25023. In the Review and Prototype Refinement stage, the researchers correct the problems found during testing, such

as sensor calibration drift and timing issues, and improve the control logic before testing the prototype again.

The expected output is the S.A.L.I.M.A.N.G.O. (Solar Automated LoRa-Integrated Mud Crab Aeration Network with Crab Grid Observation) prototype. The system monitors water quality continuously and starts aeration automatically when dissolved oxygen drops to a dangerous level, even when the connection to the house or the internet is lost. The farmer can view readings and send commands through the dashboard. The feedback loop returns the results of testing and evaluation to the input and process stages so that the prototype can be improved. Through this approach, the study aims to help farmers move from reactive, manual monitoring toward proactive, data-informed mud crab fattening.

Statement of the Problem

This study aims to solve the problem of high mortality rates and limited productivity in mud crab aquaculture, which are primarily driven by undetected water quality fluctuations, cannibalistic behavior, and the lack of off-grid monitoring capabilities for smallholder farmers. The researchers seek to answer the following:

1. What are the current operational status, management practices, and technological challenges in mud crab fattening systems among rural farmers?
2. What are the stages of development of the S.A.L.I.M.A.N.G.O. (Solar Automated LoRa-Integrated Mud crab Aeration Network with Crab Grid Observation) system using the prototyping method?
3. What is the level of acceptability of the S.A.L.I.M.A.N.G.O. system, evaluated against the ISO/IEC 25010:2011 product quality model with questionnaire items adapted from the quality measures of ISO/IEC 25023:2016, and applied to both the hardware prototype and the web dashboard, considering the following quality characteristics:

3.1. Functional Suitability;

3.2. Performance Efficiency;

3.3. Usability;

3.4. Reliability; and

3.5. Maintainability?

How does the developed system perform in terms of:

4.1. accuracy of the dissolved oxygen, pH, temperature, and salinity sensors when compared with reference standards under bench conditions and with reference measurements under pond conditions;

4.2. response time of the automated aeration, measured from the detection of dissolved oxygen below the critical threshold to the activation of the aerator;

4.3. reliability of the point-to-point LoRa telemetry between the pond float and the house receiver, in terms of packet delivery ratio, received signal strength (RSSI), and signal-to-noise ratio (SNR); and

4.4. fail-safe operation during loss of the communication link, loss of power and recovery after power restoration, and receipt of invalid commands?

Scope and Limitations of the Study

This research focuses on the development of the Solar Automated LoRa-Integrated Mud crab Aeration Network with Crab Grid Observation (S.A.L.I.M.A.N.G.O.), a smart aquaculture system designed to address the primary causes of mud crab mortality during fattening in resource-limited settings. The study encompasses the engineering of an

off-grid, solar-powered architecture equipped with a submerged multiparameter sensor hub that continuously measures dissolved oxygen (DO), temperature, pH, and salinity. The system also estimates ammonia toxicity risk by computing the toxic un-ionized ammonia (NH_3) fraction from the measured pH and temperature. It does not measure the actual NH_3 concentration, because an in-situ ammonia sensor is outside the project's budget. Intended for deployment in remote coastal or mangrove areas, the scope includes a point-to-point LoRa link at 915 MHz, using EBYTE E22-900T22D modules, for long-range, low-power data transmission between the pond float and a receiver at the farmhouse. The receiver connects to a web dashboard application for real-time monitoring and remote commands. An ESP32 microcontroller on the float executes all emergency decisions locally. Aeration runs continuously by default and is forced on whenever DO falls below a critical threshold, even when the link to the house or the internet is lost. Furthermore, the physical scope involves eight individual slotted boxes on a floating platform to prevent cannibalism.

Intended for testing in off-grid aquaculture environments, the float is supplied during testing by a protected 220 VAC line from the household's existing solar installation, which provides continuous 24-hour power through its inverter and battery bank, and the supply is stepped down to 12 VDC on the float. Because no local battery buffer is installed on the float, the AC feed constitutes a single point of failure: an interruption at the shore-end breaker or a damaged conductor halts aeration. The design mitigates this through a heartbeat-loss alarm raised at the house bridge within 90 seconds of the last received uplink, rather than through on-site energy storage. The study is also limited by the low data rate of point-to-point LoRa telemetry in the 915 MHz band. The link carries only compact, periodic frames of sensor readings, actuator states, and heartbeat signals, and it does not support continuous or high-volume data such as images or video. Its range depends on antenna placement, terrain, and vegetation, and it is evaluated only over the link distance at the test site. Salinity is measured through electrical conductivity using a K=1 probe with

a supported range of 0–20 mS/cm, or approximately 12 ppt, so salinity above this level cannot be measured accurately. The low-cost DO, pH, and salinity probes also require periodic calibration and are subject to calibration drift, and their accuracy is evaluated only for the duration of the test period. Furthermore, the system is characterized as a localized solution for the fattening of mud crab (Scylla serrata) in individual floating boxes at a single test site. It excludes hatchery, nursery, and grow-out culture, as well as indoor, vertical, and recirculating aquaculture infrastructure. The system does not include cameras or computer vision, automated feeding, or on-float energy storage. Finally, the critical DO threshold is set according to SEAFDEC/AQD water quality targets for mud crab fattening, so the system cannot be applied to other aquaculture species without recalibration.

Significance of the Study

The development of S.A.L.I.M.A.N.G.O. (Solar Automated LoRa-Integrated Mud crab Aeration Network with Crab Grid Observation) will be beneficial for the following:

For Smallholder Mud Crab Farmers. Enables real-time monitoring of water quality and automated aeration to prevent mortality events, while reducing daily labor requirements and operational costs through solar-powered, off-grid operation.

For Fishing Communities and Coastal Residents. Creates new employment opportunities in technical services and stimulates local economic activity while increasing community food security through consistent year-round mud crab production.

For Aquaculture Industry Players and Agricultural Entrepreneurs. Provides a replicable technology platform that improves production consistency and enables access to premium markets through verifiable quality and sustainability credentials.

For Government Agencies. Supports national food security and rural economic development objectives while providing evidence-based data for aquaculture policy development and environmental protection initiatives.

For the National Economy. Increases domestic aquaculture production and export earnings while creating rural employment and reducing government expenditure on disaster relief and agricultural support programs.

For Researchers and Academic Institutions. Provides a field-validated IoT platform for multidisciplinary research on smart aquaculture, sensor integration, and technology adoption in developing country contexts.

For the Proponents. Validates technical competency in IoT systems integration, embedded systems programming, and real-time data processing through successful implementation of a functional field-deployed system that generates publishable research output.

Definition of Terms

Ammonia Toxicity-Risk Estimate. The proportion of total ammonia present in the toxic un-ionized form (NH_3), computed from measured pH and temperature. In this study, it indicates relative toxicity risk, not the actual ammonia concentration in the water.

Automated Actuation. The operational mechanism that translates real-time sensor readings into physical corrective actions without constant human oversight. In this study, it involves switching air pumps and bilge pumps through a relay module, including forcing aeration on when dissolved oxygen falls below the critical threshold.

Cannibalism. A primary biological challenge and cause of mortality in mud crab aquaculture, in which crabs attack and consume one another, especially newly molted individuals. The developed system prevents this behavior by isolating each crab in an individual slotted box on a floating platform.

Cyber-Physical Systems Theory. A theoretical foundation positing that complex agricultural environments can be optimized when physical and biological processes are integrated with computational algorithms and automated networks.

Dissolved Oxygen (DO). The amount of oxygen dissolved in water, expressed in milligrams per liter (mg/L). It is the primary parameter that triggers the system's emergency aeration.

Electrical Conductivity (EC). The ability of water to conduct electric current, which increases with dissolved salt content. In this study, salinity is derived from EC measured by a K=1 conductivity probe.

Fail-Safe Operation. A control behavior in which any fault leaves the system in automatic mode with aeration on. Such faults include power-up, reset, loss of the communication link, and an invalid or expired command.

Fattening. The short-term culture of lean or newly molted mud crabs until they gain meat and reach market quality.

Heartbeat Signal. A periodic status message that the float controller sends to the house bridge every 60 seconds. If no heartbeat arrives within 90 seconds, the house bridge raises an alarm.

Internet of Things (IoT). A network of physical devices equipped with sensors, controllers, and communication modules that collect and exchange data. In this study, it enables the real-time, remote tracking of dissolved oxygen, pH, temperature, and salinity.

ISO/IEC 25010 and ISO/IEC 25023. ISO/IEC 25010 is the international product quality model that defines quality characteristics such as functional suitability, performance efficiency, usability, reliability, and maintainability. ISO/IEC 25023 defines measurement functions for evaluating those characteristics.

LoRa (Long Range, Point-to-Point). A physical-layer radio modulation technology that enables direct wireless communication between two nodes at 915 MHz, without a LoRaWAN gateway, join server, or network infrastructure. In this study, it carries low-power telemetry and commands between the pond float controller and the house bridge node over a target range of 1 to 3 kilometers line of sight.

Mud Crab Aquaculture. The cultivation of Scylla species, commonly practiced in the tropical coastal regions of Southeast Asia, India, and the Pacific Islands.

Multiparameter Sensor Array. The set of submerged sensors on the float that continuously measures dissolved oxygen, pH, temperature, and salinity.

Prototyping Model. A development method in which a working prototype is built, tested, and refined in repeated cycles until it meets its requirements. In this study, it consists of five stages: Requirements Identification and Analysis, Architectural and Quick Design, Prototype Construction and Assembly, Prototype Testing and Evaluation, and Review and Prototype Refinement.

S.A.L.I.M.A.N.G.O. An acronym for Solar Automated LoRa-Integrated Mud crab Aeration Network with Crab Grid Observation. It is the off-grid, smart aquaculture prototype developed in this study to support proactive, data-driven mud crab fattening.

Socio-Technical Systems Theory. A theory stating that a technological solution will only succeed if its infrastructure is optimized for the social and economic constraints of its target users. In this study, it justifies designing an affordable, off-grid system for smallholder farmers lacking reliable electricity and internet.

CHAPTER 2:

This section of the study discusses various studies and works related to mud crab aquaculture management, cannibalism prevention, and water quality dynamics affecting crab survival. It further examines studies on the development of smart aquaculture monitoring systems, the application of multiparameter sensor arrays for real-time water quality tracking, and the implementation of solar-powered, off-grid architectures that utilize LoRaWAN telemetry for long-range, low-power data transmission in remote coastal environments.

Mud Crab Aquaculture Management and Monitoring Challenges (SOP 1)

- **Cannibalistic Behavior and Cage Culture Practices**

  The commercial viability of mud crab (Scylla species) farming is heavily constrained by biological vulnerabilities, particularly aggressive intra-species cannibalism. Apine et al. (2023) identified territorial aggression and limited seed supply as the primary obstacles preventing sustainable production in tropical coastal communities. Furthermore, Nivas et al. (2023) emphasized that high stocking densities in unsegmented systems significantly exacerbate this cannibalism and facilitate rapid disease transmission, ultimately leading to high mortality rates and reduced farm profitability. This cannibalistic behavior becomes especially detrimental during the biological molting phases of the crabs. Examining habitat conditions and physiological vulnerabilities, Khatun et al. (2023) observed that newly molted, soft-shelled crabs are particularly defenseless against their hard-shelled counterparts. Because of this inherent biological risk, relying on open, communal farming environments invariably results in severe stock depletion, making the physical separation of individual crabs an absolute necessity to preserve the yield. The consequences of failing to implement such physical separation are severely magnified in advanced aquaculture setups. On an industrial scale, Wang et al.

(2024) demonstrated that attempting high-density temporary rearing without the use of individual compartments or physical separation directly causes severe stock loss. Beyond direct predation, this unmitigated aggression and subsequent mortality accelerate water quality degradation, creating a compounding cycle of environmental deterioration and further stock depletion. To counteract these biological and environmental risks, modern aquaculture management is shifting toward individualized cage culture practices integrated with smart monitoring. To enable non-invasive, individual-level monitoring within these segmented cage setups, Chen et al. (2023) developed the GMNet-YOLOv4 deep learning model. Their study proved that lightweight computer vision can effectively track individual crab behaviors and physical conditions in real-time, even in resource-constrained environments, ensuring that isolated crabs can be monitored efficiently without manual handling.

- **Water Quality Fluctuations and Mortality Risks**

  Physicochemical instability in marine environments represents another major cause of chronic mortality in aquaculture operations, making continuous environmental surveillance a critical necessity. In a systematic bibliometric review of IoT trends in aquaculture, Flores-Iwasaki et al. (2025) identified pH, temperature, and dissolved oxygen as the core universal indicators required to evaluate aquatic environmental stability and ensure species survival. Expanding on this baseline, Lindholm-Lehto (2023) highlighted that the continuous tracking of these metrics alongside salinity and oxidation-reduction potential is essential to prevent systemic physiological stress and the outbreak of diseases in closed or semi-closed culture systems.

  Because unseen fluctuations in these parameters can rapidly degrade the aquatic habitat, delayed detection often results in catastrophic stock losses. Addressing this urgency, Bakar et al. (2025) underscored the critical necessity of deploying

real-time, multi-parameter monitoring systems. Their research established that pairing continuous tracking with rigorous sensor calibration is the most effective method for detecting hazardous aquatic anomalies, allowing farmers to intervene before water quality degradation leads to widespread mortality.

While the need for precision monitoring is clear, the high cost of commercial-grade instrumentation frequently prevents smallholder farmers from adopting these technologies. To ensure cost-effective deployment without sacrificing reliability, Jais et al. (2024) demonstrated that statistical calibration techniques applied to low-cost sensors can yield highly dependable results. Their study proved that budget-friendly sensor arrays can achieve measurement accuracies between 76% and 97% compared to professional laboratory instruments, making real-time environmental tracking economically viable for rural aquaculture operations.

Ultimately, the effectiveness of any smart aquaculture framework relies on the integrity of the information it collects. Eneh et al. (2023) proved that improving raw sensor data quality directly enhances the predictive reliability of downstream machine learning algorithms and automated decision-making processes. By establishing clean, highly accurate water quality data as a prerequisite, these IoT systems enable a reliable foundation for the automated control mechanisms necessary to maintain optimal water conditions and protect crab yields.

- **Reactive vs. Proactive Farm Management Approaches**

The modern aquaculture industry is fundamentally transitioning from traditional, reactive practices toward proactive and predictive architectures. Lujan (2026) notes that reactive systems attempt to correct environmental damage only after it has occurred, whereas predictive systems anticipate water quality crises before they manifest, utilizing automated aeration and filtration to prevent lethal toxicity. This shift addresses the severe limitations of traditional farm management, which relies

on periodic manual sampling that frequently misses early signs of deteriorating water quality. Consequently, hazardous anomalies are detected too late for effective intervention. The severe economic and biological consequences of this delay were documented in the Philippine context by Asis and de la Cruz (2026), whose research highlighted that reactive management directly exacerbated catastrophic mass mortality incidents across major lakes between 2023 and 2024. Validating this urgency globally, Lazar et al. (2026) explicitly stated that effective ecological management requires continuous, preventive strategies rather than post-degradation reactive measures. Dam et al. (2026) reinforced that the core bottleneck in contemporary aquaculture is no longer sensor availability, but the ability to translate environmental intelligence into timely, reliable, and closed-loop physical actions to manage farms proactively. To counteract these fatal delays, recent deployments champion continuous telemetry paired with anticipatory trigger mechanisms. Pines et al. (2026) demonstrated that relying solely on a reactive dissolved oxygen threshold is flawed, as rising water temperatures naturally degrade oxygen solubility before probes register a critical drop. To solve this, their system utilized a proactive dual-threshold logic that automatically activated aerator pumps when dissolved oxygen fell below 6.5 mg/L or when temperatures exceeded 29°C, stabilizing the aquatic environment before hypoxic stress could occur. The most reliable form of this proactive management is complete closed-loop automation, where embedded microcontrollers autonomously command physical actuators without constant human oversight. Guansing et al. (2026) proved the efficacy of this approach in intensive Philippine aquaculture setups by utilizing fuzzy logic to instantly correct imbalances across parameters like pH, ammonia, turbidity, and total dissolved solids, significantly improving survival rates over conventional manual techniques. Furthermore, Dam et al. (2026) observed that such proactive, data-driven farms experience up to a 30% reduction in feed waste and a 20% decrease in mortality rates. Collectively, these studies strongly validate the

integration of continuous multiparameter sensing and automated actuation to physically correct the aquaculture ecosystem before mass mortality events unfold, directly supporting the dynamic mechanisms proposed in the S.A.L.I.M.A.N.G.O. framework.

- **Constraints of Off-Grid, Rural Coastal Aquaculture Operations**

  The transition toward modernized aquaculture is heavily impeded by the infrastructural realities of remote coastal environments. Subowo and Pradita (2025) noted that while Internet of Things (IoT) technologies offer transformative potential for real-time environmental monitoring, most existing solutions are tailored for commercial-scale operations and demand expensive hardware, sophisticated data infrastructure, and continuous internet connectivity. Consequently, these stringent requirements render modern IoT systems highly impractical for small-scale fish farmers operating in remote, off-grid areas with limited power and network availability. To bridge this connectivity gap, Baena-Navarro et al. (2025) emphasized that monitoring systems in rural regions must utilize adaptable, low-cost technologies and localized networks that can function efficiently without relying on high-speed internet. Beyond technological barriers, off-grid coastal setups are continuously exposed to severe environmental and climatic vulnerabilities. Sumalpong and Embornas (2024) observed that unpredictable weather patterns, such as the "Kulyada" phenomenon, bring shifting winds and extreme conditions that directly disrupt fishing activities and exacerbate the vulnerabilities of small-scale fisheries. Similarly, Densing and Matutes (2025) highlighted that prolonged dry spells, extreme temperature fluctuations, and heavy rainfall in fragile wetland and coastal ecosystems lead to rapid water quality degradation, significantly threatening aquaculture sustainability. In the context of mud crab (genus Scylla) farming, Murak et al. (2026) demonstrated that such habitat instability, combined with organic matter accumulation and limited water exchange, increases host stress and severely

elevates the risk of catastrophic pathogen outbreaks, including Vibrio infections and White Spot Syndrome Virus (WSSV). Because rural farmers generally lack real-time surveillance, Murak et al. (2026) further explained that these hidden anomalies often go undetected until significant stock losses and mass mortality have already occurred. The operational difficulties in off-grid farming are further compounded by entrenched socio-economic limitations. Densing and Matutes (2025) pointed out that small-scale producers typically operate under severe capital constraints and lack access to formal financial services, which restricts their ability to invest in resilient pond infrastructure, professional-grade sensors, or quality formulated feeds. This financial insecurity forces farmers to rely heavily on subsistence-level, manual management practices. Murak et al. (2026) emphasized that the labor-dependent nature of routine farm operations—such as manual feeding, water quality monitoring, and harvesting—limits efficiency gains and scalability, trapping farmers in a reactive and high-risk production cycle. Furthermore, Azcuna-Montaño and Largo (2026) indicated that high operational costs, perceived financial risks, and a lack of institutional support often hinder the widespread adoption of sustainable and integrated aquaculture technologies among marginalized coastal communities.

Accuracy and Efficiency of Multimodal Aquaculture Sensing and Control (SOP 4)

- **Water Quality Parameter Detection (Dissolved Oxygen, pH, Temperature, Ammonia, Salinity)**

The accuracy and reliability of multiparameter water quality sensing form the technical foundation of any smart aquaculture system, and recent studies have focused on improving both the affordability and precision of these sensor networks. Eneh et al. (2023) developed an improved Internet of Things (IoT) sensor system for a mobile aquaponics setup that continuously collected temperature, pH, dissolved oxygen, turbidity, ammonia, and nitrate data across twelve fish ponds. Their refined system produced a labeled dataset

that, when used to train machine learning models for predicting catfish growth, achieved a substantially lower prediction error than the earlier version of the system, demonstrating that improving raw sensor data quality directly enhances the reliability of downstream decision-making algorithms.

A broader perspective on sensor adoption trends is offered by Flores-Iwasaki et al. (2025), who conducted a systematic review and bibliometric analysis of 217 IoT-based water quality monitoring studies published between 2020 and 2024 across biofloc, recirculating, and aquaponic systems. Their findings revealed that pH sensors were used in 98.2 percent of the reviewed studies, followed by temperature sensors at 92.9 percent and dissolved oxygen sensors at 62.5 percent, confirming that these three parameters represent the technological baseline for aquaculture monitoring systems and reinforcing the parameter selection adopted in the mud crab fattening system's sensor array.

Addressing the cost barrier associated with commercial-grade sensors, Zuhaer et al. (2025) designed and field-tested a low-cost IoT-integrated water quality monitoring device for tilapia tanks featuring in-house developed dissolved oxygen and ammonia sensors alongside commercially available pH, temperature, and turbidity probes. The system achieved an accuracy exceeding 98 percent for dissolved oxygen and above 99 percent for ammonia readings while reducing overall sensor package costs by approximately 85 percent, illustrating a viable pathway for delivering laboratory-grade accuracy to smallholder farmers operating under tight budget constraints.

Similarly concerned with the accuracy of budget sensors, Jais et al. (2024) evaluated an IoT-based water quality monitoring system for Asian seabass aquaculture tanks by calibrating low-cost sensors against a professional-grade YSI Pro instrument using simple linear regression. Following calibration, the system achieved accuracy levels ranging from 76 to 97 percent with relative error values as low as 0.27 percent, demonstrating that

statistical calibration techniques can meaningfully narrow the performance gap between low-cost and professional water quality instrumentation over sustained field deployment.

Finally, Lindholm-Lehto (2023) provided a comprehensive review of water quality monitoring practices in recirculating aquaculture systems, identifying pH, temperature, oxidation-reduction potential, turbidity, salinity, and dissolved oxygen as the core parameters requiring continuous surveillance to prevent stress and disease in cultured species. The review further emphasized that Internet of Things-based systems paired with artificial intelligence enable real-time measurement and early warning of critical water quality deviations, supporting the theoretical justification for pairing continuous multiparameter sensing with automated corrective actuation, as proposed in the present study.

- **Vision-Based Biological Monitoring (Molting Status, Predator Detection)**

Beyond physicochemical sensing, computer vision has emerged as a critical tool for monitoring the biological status of cultured crustaceans without requiring constant manual inspection. Chen et al. (2023) proposed GMNet-YOLOv4, a lightweight crab detection and gender classification model built on a GhostNet backbone, designed specifically to address the challenge of identifying individual Chinese mitten crabs within images containing complex backgrounds and multiple overlapping specimens. Their model reduced computational parameters while maintaining detection accuracy, demonstrating that lightweight deep learning architectures can support real-time biological monitoring on resource-constrained embedded hardware suitable for field deployment.

Focused specifically on crab health and survival status, Feng et al. (2023) developed a fusion recognition algorithm combining YOLOv5 and RefineContourNet (RCN) to assess the survival rate of Portunus crabs in single-frame three-dimensional aquaculture. The system analyzed multiple visual parameters, including residual bait ratio, centroid movement distance, and body rotation angle, which were subsequently integrated through local

weighted linear regression and a softmax classification model, enabling automated, non-invasive assessment of crab vitality directly from captured video footage.

Directly addressing molting detection, Saputra et al. (2025) applied YOLO-based object detection models, including YOLOv8 and YOLOv11, to automatically identify molting crabs, shed crab skins, and newly molted crabs concealed within box baskets that typically obstruct camera visibility during soft-shell crab production. Their trained models achieved detection accuracies between 96 and 98 percent, substantially outperforming manual visual inspection methods and confirming that automated molting detection can eliminate the labor-intensive, time-sensitive monitoring traditionally required to harvest soft-shell crabs before their new exoskeleton hardens.

In a related effort, Zhang et al. (2024) introduced YOLOF, an improved detection algorithm built upon YOLOv5 that incorporates a FasterNet feature extraction backbone and an additional auxiliary prediction layer to enhance soft-shell crab detection accuracy. The researchers also constructed the Crab Molting Detection Dataset (CMP) to address the scarcity of publicly available annotated datasets for this task, providing a valuable benchmark resource for subsequent vision-based molting detection research in crustacean aquaculture.

On the predator detection front, Bose et al. (2024) designed an integrated pond management system combining IoT-based water quality monitoring with a passive infrared (PIR) motion-sensing bird deterrent mechanism for fish farms. Upon detecting avian predator movement near the pond, the system automatically triggered a water-spraying mechanism to repel birds without the need for costly or potentially harmful net enclosures, illustrating a practical, non-invasive model for automated predator detection and deterrence that parallels the localized vision-based predator monitoring functions envisioned for the crab cage system.

- **Automated Actuation Response Mechanisms (Aerators, Dynamic Pumps)**

Automated actuation forms the operational backbone of any proactive aquaculture system, translating sensor readings into physical corrective action without requiring constant human oversight, and recent studies have validated this approach across different water quality parameters. Ijaz and Iqbal (2022) developed a solar-powered water pumping automation system for an off-grid fish farm in rural Pakistan, using an Arduino Uno microcontroller paired with an ultrasonic water level sensor to switch the pump on and off automatically. Their design demonstrated that microcontroller-based actuation can eliminate the need for on-site personnel while remaining inexpensive enough for smallholder adoption, directly supporting the socio-technical justification for automated pump control in remote farming contexts.

Focusing specifically on salinity regulation, Putri et al. (2026) designed an Arduino-based automatic water salinity control system using a Gravity TDS sensor and a closed-loop relay mechanism that activates one of two water pumps, a freshwater pump to dilute excess salinity and a saltwater pump to raise it, whenever readings cross predefined thresholds. Their system achieved a root mean square error of only 1.902 ppm and a mean absolute error of 1.76 ppm relative to reference instruments, confirming that threshold-triggered dynamic pumping can reliably restore salinity to a target setpoint even under continuously changing mixing conditions, a finding that closely parallels the dynamic pump mechanism proposed for correcting rainfall-driven salinity drops in mud crab cages.

Extending automated actuation to multiple parameters simultaneously, Meza-Solis et al. (2025) implemented an Arduino Mega-based control system for industrial aquaculture that continuously compares temperature and pH sensor readings against target ranges and triggers electronic actuators to adjust conditions without manual intervention. The study emphasized that such automated regulation reduces workers' reliance on reactive interventions and can be scaled to larger production environments, reinforcing the viability of embedded, low-cost actuation logic for commercial-grade water quality correction.

Addressing dissolved oxygen management specifically, Suriasni et al. (2024) integrated an IoT-based aerator on/off control system into a laboratory-scale recirculating aquaculture system, automatically activating aeration whenever dissolved oxygen fell outside the 5–7.8 mg/L range required for optimal biofilm nitrification. Their results showed that sustained automated aeration control allowed the system to achieve a 50 percent ammonia removal efficiency, demonstrating that timely, sensor-triggered aerator activation has downstream benefits beyond oxygen maintenance alone, including improved water chemistry stability that reduces mortality risk.

Most directly relevant to the present system's cage-based deployment, Sangkaphet et al. (2026) developed and field-evaluated an IoT- and LoRa-based water-quality-monitoring and aeration-actuation system for open-water tilapia cage farming, in which distributed control nodes automatically triggered oxygen-pump actuation upon detecting unsafe dissolved oxygen levels. Detection-to-actuation latency averaged only 208.6 to 289.7 milliseconds for single-hop communication, and farmer satisfaction assessments indicated strong practical acceptance of the automated aeration response, providing direct empirical support for combining long-range wireless connectivity with rapid, automated aerator triggering in submersible cage systems similar to those proposed in this study.

- **LoRaWAN Network Telemetry Performance and Reliability**

The reliability of LoRaWAN as a long-range, low-power telemetry backbone for remote aquaculture monitoring has been increasingly validated through field trials in constrained and infrastructure-poor environments. Chapungo and Postolache (2025) conducted an experimental evaluation of LoRaWAN connectivity in a rural area of Mozambique with no existing electricity or mobile network infrastructure, deploying an Arduino-based node powered solely by a power bank. Their trials achieved a packet delivery rate above 89 percent and sustained autonomous operation for over 24 hours, while also confirming that terrain elevation and distance significantly affect signal strength and packet

reception, underscoring the importance of gateway placement in coastal or mangrove deployments comparable to mud crab farming sites.

Applying LoRaWAN directly to aquaculture water quality regulation, Espena et al. (2023) built a shrimp farm monitoring and control prototype that transmits temperature, pH, salinity, and dissolved oxygen readings from a centrally located sensor node and automatically triggers corrective actuation when parameters deviate from safe ranges. Their system demonstrated that LoRaWAN connectivity can support both real-time farmer notification and automated response within a single low-power architecture, validating the integration of long-range telemetry with actuation logic in a brackish water culture setting similar to mud crab farming.

Reinforcing the feasibility of solar-autonomous LoRa telemetry, Bullecer et al. (2025) developed an energy-autonomous water quality monitoring node combining an ESP32-LoRa module with photovoltaic charging and a LiFePO4 battery buffer, enabling continuous operation without grid access. Sensor readings relayed through the LoRa link showed close agreement with reference instruments, with percentage differences under 0.5 percent for temperature, confirming that solar-powered LoRa nodes can maintain both measurement accuracy and transmission reliability over sustained, unattended deployment.

Investigating the structural limits of single-gateway LoRaWAN coverage, Durand and Booysen (2025) proposed and simulated a complementary mesh-topology LoRa network designed to extend connectivity beyond a single gateway's range. Their results showed that packet delivery ratio for nodes located more than 5.8 kilometers from a centralized gateway improved from an average of 40.2 percent to 73.78 percent when mesh routing was applied, highlighting a practical pathway for maintaining telemetry reliability across the large, dispersed cage layouts typical of coastal mud crab farms that may exceed a single gateway's effective range.

Finally, Alsewari and Ali (2024) developed a LoRaWAN-based IoT water quality monitoring system for rural deployment and evaluated its data transmission consistency across an extended low-power sensor network integrated with cloud-based platforms for real-time analysis. Their findings emphasized that careful network configuration and gateway architecture are essential for maintaining stable, uninterrupted data flow in rural, connectivity-poor regions, further supporting the telemetry design assumptions underpinning the proposed system.

System Development and Prototyping Approaches for Smart Aquaculture (SOP 2)

- **CDIO Framework and Prototyping Methodologies**

The application of structured prototyping methodologies is essential in bridging theoretical engineering concepts with functional field deployments. Habbal et al. (2024) emphasized that applying structured prototyping frameworks is critical in addressing complex human and environmental challenges, arguing that education and social connection are indispensable to producing engineers capable of tackling open-ended, real-world problems. Their work compiles case studies drawn from the United Nations Sustainable Development Goals alongside curriculum-change experiences at Harvard University, the University of Technology Sydney, and Aalborg University, offering a structured comparison of techniques for managing complexity in engineering education. This is particularly relevant in time-constrained capstone projects where the agile integration of hardware and software is necessary. For instance, García-Ávila et al. (2026) demonstrated how a CDIO-based (Conceive-Design-Implement-Operate) framework could compress the full-cycle development of complex electromechanical prototypes into a single academic semester. Their framework combined Design for Manufacturing, design thinking, and project-based learning through weekly sprints, individual logbooks, a live bill of materials, rolling-wave Kanban-Gantt planning, gated reviews, and an early hard-point freeze, and was validated with 26 mechanical engineering undergraduates developing a Shell Eco-marathon

competition vehicle within a 15-week delivery window while maintaining control of scope, schedule, and cost. By adopting these agile frameworks, engineers can systematically design physical systems like S.A.L.I.M.A.N.G.O. while maintaining strict control over scope, budget, and functionality.

In the context of aquaculture, the CDIO approach ensures that the resulting prototypes are both environmentally resilient and user-centric. Olanubi et al. (2024) successfully detailed the design methodology and prototype development of an IoT-based intelligent water quality management system tailored specifically for aquaculture, noting that traditional water quality management forces farmers to rely on intuition or on periodic sample collection and laboratory testing—a process that is tedious, labor-intensive, and elongates the response time needed to act on deteriorating conditions. Building on this, Ivanov et al. (2026) designed an autonomous hydrochemical monitoring prototype utilizing low-power LoRaWAN nodes to ensure continuous, energy-efficient operation in large freshwater facilities, directly addressing the need for smart monitoring within the broader Agriculture 4.0 movement. Their sensor modules perform on-board signal preprocessing, including anomaly detection and short-term forecasting of key hydrochemical parameters, while an Extended Kalman Filter fuses heterogeneous sensor data with predictive estimates to increase measurement reliability. However, technical functionality must be paired with usability for rural users. Adhi and Irianto (2026) validated this by evaluating a smart farming application prototype, stating that systems achieving high System Usability Scale (SUS) scores "tend to achieve better user acceptance and operational efficiency" among farmers. Their own prototype, tested with ten farmer respondents aged 23 to 70, achieved a SUS score of 86 against a global benchmark of 68, while all five core functional features passed Black Box testing without error. This highlights the necessity of human-centered evaluation in deploying off-grid aquaculture prototypes, where digital literacy cannot be assumed.

- **Cyber-Physical Systems Integration in Aquaculture**

Cyber-Physical Systems (CPS) theory forms the technological backbone of modern smart aquaculture by tightly coupling computational algorithms with physical environmental processes. Roy et al. (2024) explored the large-scale integration of Artificial Intelligence and CPS, demonstrating how automated environmental regulation fundamentally revolutionizes aquaculture sustainability by minimizing manual labor across tasks such as fish species classification, health monitoring, feed regulation, and water quality management. This transition towards "Fish Farming 5.0" heavily relies on integrating advanced tools—such as environmental sensors and machine vision—into a cohesive cyber-physical architecture to manage operations proactively rather than reactively. By linking biological observation with real-time data processing, a CPS framework transforms passive monitoring into active environmental management, a core requirement for mitigating mortality in vulnerable species like mud crabs.

The practical deployment of these cyber-physical systems depends on seamless closed-loop control mechanisms. To manage complex environmental variables effectively, Al-Mutairi and Al-Aubidy (2023) demonstrated the efficacy of embedding fuzzy logic directly into the cyber-physical control loop of an aquaculture IoT platform, allowing the system to independently evaluate and respond to changing pond conditions. Their architecture treated each fish pond as a node within a wireless sensor network, with an embedded microcontroller connected to a set of sensors and actuators, drawing on five in-pond sensors and three environmental sensors per node and two dedicated fuzzy controllers—one for water quality and one for the surrounding environment—to achieve measurement accuracy comparable to commercial monitoring devices. These integrations heavily justify the S.A.L.I.M.A.N.G.O. system's use of automated aerators and dynamic pumps as physical actuators driven by intelligent software telemetry, since they demonstrate that a fully closed sense-evaluate-actuate loop can operate reliably without constant human oversight even under the variable conditions typical of open-water and semi-controlled culture environments.

- **Socio-Technical Design for Off-Grid Farming Communities**

Deploying smart aquaculture systems in rural environments requires a socio-technical approach that respects the infrastructural constraints of off-grid coastal communities. Chapungo and Postolache (2025) highlighted the structural limits of deploying IoT infrastructure in areas lacking electricity or mobile networks, empirically evaluating LoRaWAN connectivity in remote Mozambican villages to prove its viability under extreme constraints. Their field trials used an Arduino-based node with a small antenna powered solely by a power bank, collecting signal strength, signal-to-noise ratio, and packet delivery data at multiple georeferenced points along a one-kilometer path; the network sustained a packet delivery rate above 89 percent and operated autonomously for over 24 hours, while also confirming that terrain elevation and distance significantly affect signal quality. Because stable internet is rarely available in these coastal areas, Jabbar et al. (2024) emphasized that "careful network configuration and gateway architecture are essential for maintaining stable, uninterrupted data flow" when deploying LoRaWAN-based water quality monitoring systems in connectivity-poor rural deployments. Their solar-powered, waterproof system paired a LoRa node carrying pH, total dissolved solids, turbidity, and temperature sensors with an Arduino microcontroller and LoRa shield, relaying data through a LoRaWAN gateway to cloud dashboards for continuous remote access, and was field-validated at Gambang Lake against laboratory sample analysis. This underscores the necessity of choosing low-power, long-range telemetry over traditional cellular networks to ensure system reliability for rural farmers.

Beyond mere connectivity, socio-technical design must account for power autonomy and economic accessibility. Bullecer et al. (2025) addressed this by developing a highly relevant energy-autonomous, solar-powered LoRa system specifically designed for remote water quality monitoring, combining an ESP32-LoRa module with photovoltaic charging and a LiFePO4 buffer battery to enable continuous operation without grid access; sensor readings calibrated against NIST-traceable and standard reference solutions showed

percentage differences under 0.5 percent for temperature, confirming that solar autonomy need not come at the cost of measurement accuracy. Furthermore, Soto-Rios et al. (2023) discussed cross-sector socio-technical innovation processes within the water and wastewater sector of the Latin American region, arguing that responsible resource management in developing areas must align with local capabilities and territorial inequalities rather than assuming uniform infrastructure access across a country or region. While their focus is circular-economy governance rather than aquaculture specifically, the underlying principle—that technological solutions must be calibrated to the uneven distribution of resources, skills, and infrastructure across a target population—applies directly to the deployment of off-grid monitoring systems among smallholder farmers. Therefore, systems like S.A.L.I.M.A.N.G.O. must balance advanced computational capabilities with the off-grid, low-cost realities of their end-users to achieve true agricultural modernization.

- **Hardware–Software Integration in Embedded IoT Systems**

The operational success of an aquaculture monitoring system heavily depends on the robust integration of embedded hardware sensors with cloud-based software telemetry. Zuhaer et al. (2025) detailed the seamless embedded integration of low-cost, in-house developed hardware sensors—specifically for dissolved oxygen and ammonia—with IoT software dashboards to achieve sustainable, real-time water quality monitoring, reporting accuracy exceeding 98 percent for dissolved oxygen and approaching 99 percent for ammonia readings while offering real-time updates through a companion mobile application. By directly embedding multiparameter monitoring methods into the hardware layer, developers can ensure uninterrupted and effective data transmission over IoT platforms (Shete et al., 2024), whose Arduino Uno-based architecture integrated a calibrated pH probe and dissolved oxygen sensor to deliver comprehensive, real-time water quality data intended to support proactive rather than reactive farm management. This hardware-software synergy guarantees that physical environmental changes are instantaneously and accurately reflected on the farmer's remote monitoring interface.

In addition to sensing, effective hardware-software integration governs the automated actuation of corrective mechanisms. Sarwar and Iqbal (2022) established the foundational integration of microcontroller software logic with solar-powered pump hardware, demonstrating that embedded systems can operate autonomously in off-grid aquaculture without human intervention; their Arduino Uno microcontroller, paired with an ultrasonic water level sensor, switched a solar-powered pump on and off automatically for a rural fish farm in Pakistan with no reliance on grid electricity or on-site personnel. Building upon this concept, Putri et al. (2026) explained the embedded closed-loop relay mechanisms that successfully tie hardware sensors directly to physical pump actuators for automatic water salinity control, using a Gravity Total Dissolved Solids sensor to trigger either a freshwater pump to dilute excess salinity or a saltwater pump to raise it, achieving a root-mean-square error of only 1.9 parts per million against reference instruments. To maximize field reliability under unpredictable network conditions, Mnyanghwalo et al. (2024) validated the strategy of "distributing processing power directly into field-embedded nodes rather than relying entirely on central software servers" in their design of IoT-based sensor nodes for fault detection in electrical distribution networks—a parallel domain that, like aquaculture telemetry, depends on dense, geographically dispersed sensing points reporting back to a central system without the benefit of constant human supervision. Distributing this computational load ensures that automated actuation remains functional even when long-range network uplinks experience temporary latency.

System Quality and Acceptability Standards (SOP 3)

- **Functional Suitability of IoT-Based Aquaculture Systems**

Functional suitability, one of the primary characteristics of the ISO/IEC 25010 product quality model, evaluates whether a developed system provides functions that meet the stated and implied needs of its intended users. Several recent studies have applied this standard directly to IoT-based agricultural and aquaculture systems to assess how

completely and correctly the developed functions perform. Sapin et al. (2022) developed an intelligent aquaculture system integrating a geographic information system and a recurrent neural network to simulate pisciculture production in Taal Lake, Philippines, and evaluated the system using the ISO/IEC 25010 standard. The system obtained a functional suitability rating of 3.78, interpreted as “strongly acceptable,” indicating that the dashboard, simulation, and forecasting modules satisfied the operational needs of the Bureau of Fisheries and Aquatic Resources personnel who served as respondents.

Similarly, Javier et al. (2022) assessed MangngalApp, an integrated technology package developed for fisheries and aquaculture post-harvest innovations in the Philippines, using both the Technology Acceptance Model and the ISO/IEC 25010 software quality standards. Twenty technical respondents evaluated the platform across compatibility, reliability, portability, and functional suitability, with results showing that the application complied with these characteristics to a high extent, confirming that fisheries-oriented digital tools can achieve strong functional performance even when developed for resource-limited rural users.

In a closely related agricultural application, Binayao et al. (2024) developed a solar-powered, IoT-based smart water irrigation system for rice farming and evaluated it against the ISO/IEC 25010:2011 quality model. The overall system, including its moisture, humidity, and temperature detection functionalities, received an “excellent” rating with a weighted mean of 4.32, demonstrating that solar-integrated IoT systems designed for smallholder farmers can fulfill their intended functional specifications reliably under field conditions.

Focusing specifically on aquaculture pond monitoring, Erawati et al. (2025) designed an IoT-based water quality monitoring system for tilapia fish farming ponds and subjected the prototype to an expert feasibility evaluation. Three expert evaluators rated the system's functionality and feasibility for aquaculture use at an average index of 95.93 percent,

categorized as “very good,” confirming that the system's sensing and alerting functions operated according to their intended design specifications in a real fish farming environment.

Lastly, Al-Mutairi and Al-Aubidy (2023) designed and implemented an IoT-based smart monitoring and management system for fish farming that incorporated two fuzzy logic controllers to regulate water quality and pond environmental conditions using a network of five in-pond sensors and three environmental sensors per node. Their system demonstrated that embedding rule-based decision logic directly into the functional architecture of an aquaculture IoT platform enables the system to independently sense, evaluate, and respond to changing pond conditions, illustrating a functionally complete cyber-physical control loop consistent with the automated actuation goals of similarly designed aquaculture systems.

- **Performance Efficiency and Usability Evaluation**

Performance efficiency and usability are two further characteristics of the ISO/IEC 25010 model that respectively measure a system's resource utilization and response behavior, and the ease with which intended users can operate it. Several studies have quantified these characteristics for IoT-based monitoring systems using accuracy testing, cost analysis, and the System Usability Scale (SUS). Sung et al. (2023) developed an IoT-based aquaculture monitoring system for catfish ponds integrated with the Firebase real-time database and evaluated its performance through sensor accuracy testing and cost analysis. The system achieved an overall measurement accuracy of 95.51 percent relative to calibrated instruments and was found to be 94.21 percent more cost-effective than labor-based monitoring, demonstrating strong performance efficiency in both computational accuracy and resource utilization.

Addressing usability specifically, Rofiansyah et al. (2025) developed HydroFarm, an IoT-based control and monitoring system for hydroponic plant growth that combines multiparameter sensors with a convolutional neural network for plant health assessment via a mobile application. Usability testing conducted with twenty respondents using the System

Usability Scale produced an average score of 81.875, placing the application in the “excellent” and grade B classification range, which indicated that the mobile interface was well-designed, informative, and easy for urban farmers to operate despite the technical complexity of the underlying image-processing pipeline.

In a similar usability-focused evaluation, Adhi and Irianto (2026) developed a mobile-based smart farming application integrated with IoT sensors for real-time monitoring of temperature, humidity, and light intensity, and evaluated the system using both Black Box functional testing and the System Usability Scale. All five core functional features passed testing without error, and the usability evaluation involving ten farmer respondents produced a SUS score of 86, well above the global benchmark of 68, confirming that the application delivered both reliable performance and a high level of perceived ease of use among farmers with limited digital literacy.

Because network transmission speed and reliability are core determinants of performance efficiency in remote IoT deployments, Syed Taha et al. (2024) evaluated the performance of a LoRa-based wireless network integrated into a water quality monitoring system, testing signal strength, signal-to-noise ratio, and packet loss across varying antenna gains and spreading factors. Their results showed that packet loss remained below 10 percent under most tested configurations, providing empirical benchmarks for how spreading factor and antenna gain settings affect the time behavior and transmission reliability of LoRa-based telemetry in water-based monitoring deployments.

Complementing this network-level analysis in an aquaculture-specific context, a 2024 study published in the Jurnal Ilmiah Teknik Elektro Komputer dan Informatika analyzed the performance of a LoRaWAN network carrying multi-sensor data, including pH, ammonia, dissolved oxygen, salinity, and temperature readings, for a vaname shrimp farming monitoring system operating at the 920–923 MHz frequency band. The study found that the network sustained transmission bitrates between 5.5 and 21.9 kbps across varying

distances, offering practical performance benchmarks for spreading factor and antenna configuration that directly inform the design of low-power, long-range telemetry links for crustacean aquaculture systems.

- **Reliability and Maintainability of Embedded/Field-Deployed Systems**

The reliability and maintainability of embedded systems deployed in harsh, unattended field conditions determine whether an IoT solution survives beyond a laboratory demonstration, and recent studies have documented both the failure modes and the design strategies that keep such systems operational over extended periods. Royer et al. (2024) reported on four years of continuous operation of a LoRaWAN-based sensor and gateway network deployed on Mount Etna, Sicily, including a period of over 60 volcanic paroxysms; while most sensors and communicating nodes left on the exposed volcanic edifice were destroyed by extreme winter weather and eruptive activity, the gateway infrastructure situated at the foot of the volcano remained fully operational throughout, demonstrating that strategic placement of critical hardware away from the harshest exposure is central to long-term field reliability.

Addressing maintainability under resource constraints, Safi et al. (2024) designed and field-tested an LPWAN-based emergency response system for visually impaired individuals on a university campus, finding that the Heltec LoRa 32 nodes achieved 98 percent communication reliability, which the authors attributed to the technology's low-power, long-range architecture that reduces the frequency of battery replacement and node servicing in continuously operating safety-critical deployments. Similarly concerned with long-duration field stability, Jeon et al. (2025) developed WaggleNet, a LoRa- and MQTT-based dual-scope beehive monitoring system, and recorded system uptime exceeding 95 percent over a two-month deployment, with the recorded downtime traced almost entirely to master-node Wi-Fi disconnections and battery depletion rather than to LoRa communication or sensor hardware failure, illustrating that maintainability bottlenecks

in field-deployed embedded networks often lie at the power and gateway layer rather than the sensing layer itself.

Extending reliability engineering to power-grid embedded sensing, Mnyanghwalo et al. (2024) deployed IoT-based sensor nodes using a distributed processing architecture for fault detection and classification in electrical secondary distribution networks, demonstrating that distributing computation across field nodes rather than relying on centralized processing improved the practicality of continuous fault monitoring in complex, high-density networks common to developing-country infrastructure. Finally, Singh et al. (2024) conducted an exploratory analysis of reliability-designing approaches for IoT systems using a smart building management case study, showing that architectural strategies such as component redundancy, fault-tolerant control logic, and predictive maintenance scheduling significantly reduced downtime and maintenance costs compared to conventional designs, reinforcing that reliability and maintainability in embedded IoT systems are best achieved through deliberate architectural provisioning rather than as an afterthought of deployment.

- **ISO/IEC 25023 Measurement Functions in Practice**

Beyond the definitional characteristics of ISO/IEC 25010, several recent studies have operationalized the standard by applying the quantitative measurement functions defined in ISO/IEC 25023 to real, deployed systems, translating abstract quality characteristics into computable indicators. Moumane et al. (2024) evaluated the software product quality of three reproductive-health mobile applications in Morocco by pairing ISO/IEC 25010 characteristics with ISO/IEC 25023 measurement functions, developing a checklist-based impact-degree formula that quantified how strongly each application requirement influenced sub-characteristics such as functional appropriateness, reliability, and maintainability; their results showed that functional suitability requirements had the greatest measurable downstream effect on reliability and security sub-characteristics, validating the use of

25023's formula-based approach for prioritizing quality improvements in health-monitoring software.

Applying the standard to a security-sensitive government use case, Taniasari and Shodikin (2024) measured the security quality of a hospital management information system using the five ISO/IEC 25023 security sub-characteristics of confidentiality, integrity, non-repudiation, accountability, and authenticity, computing a distinct numerical value for each sub-characteristic rather than relying on a single aggregated score, which allowed the researchers to pinpoint non-repudiation as the weakest measured dimension despite an otherwise acceptable overall rating. In a related security-measurement application, Correa et al. (2022) evaluated a distributed autonomic scientific-publishing tool built on a permissioned blockchain using ISO/IEC 25023 security measures, finding that the standard's quantitative functions could be meaningfully adapted to blockchain-based distributed architectures, an environment quite different from the conventional client-server systems for which the measures were originally illustrated, indicating the flexibility of the ISO/IEC 25023 measurement approach across heterogeneous system architectures.

Demonstrating that the same measurement logic extends to embedded, solar-powered IoT hardware analogous to the present study, Sigongan et al. (2023) developed a solar-powered smart garbage-segregation bin with SMS notification and machine-learning image processing and evaluated it against the eight ISO/IEC 25010 characteristics using computed Likert-based numerical scores for each sub-characteristic, obtaining an overall weighted mean of 4.55 ("excellent"), including strong quantified scores for reliability (4.52) and maintainability (4.58); this operationalized, per-characteristic scoring approach mirrors the measurement-function logic formalized in ISO/IEC 25023 and shows its applicability to solar-powered, field-deployed embedded systems similar in architecture to our proposed system.

Synthesis of the Reviewed Literature and Studies

Recent advancements in smart aquaculture establish multiparameter water quality sensing and computer vision as the essential technological baseline for preventing disease and mortality in marine environments. Continuous tracking of dissolved oxygen, pH, temperature, ammonia, and salinity directly enhances the reliability of downstream forecasting and decision-making algorithms used in aquaculture management. To overcome the cost barriers of commercial instrumentation, studies demonstrate that statistical calibration techniques and in-house sensor designs can reduce equipment costs by up to 85% while maintaining measurement accuracy between 76% and over 99%. Complementing physicochemical detection, lightweight deep learning models and object detection architectures such as YOLOv5, YOLOv8, YOLOv11, and YOLOF enable non-invasive, real-time biological monitoring directly from captured video footage. These vision systems achieve between 96% and 98% accuracy in identifying molting crabs and shed exoskeletons concealed within aquaculture baskets, successfully eliminating labor-intensive manual inspection while integrating automated predator deterrence mechanisms without requiring costly physical enclosures.

The operational backbone of proactive aquaculture relies on automated actuation and long-range telemetry to translate real-time sensor data into rapid environmental corrections without constant human oversight. Microcontroller-based closed-loop systems reliably automate critical physical responses such as dynamic water pumping to dilute or concentrate salinity and solar-powered aeration to maintain dissolved oxygen setpoints with detection-to-actuation latencies averaging only 208.6 to 289.7 milliseconds. Sustained automated aeration not only prevents suffocation but also stabilizes water chemistry, improving ammonia removal efficiency by up to 50%. In remote, infrastructure-poor coastal environments lacking grid electricity and internet, LoRaWAN technology provides an effective low-power communication backbone capable of sustaining transmission bitrates

between 5.5 and 21.9 kbps across extended distances. Field evaluations confirm that solar-autonomous LORA nodes achieve packet delivery rates exceeding 89% and system uptimes above 95%, with complementary mesh-topology routing further improving packet delivery from 40.2% to 73.78% for nodes located beyond a single gateway's 5.8-kilometer range.

Evaluating smart aquaculture platforms against standardized quality frameworks ensures technical reliability and end-user acceptability among smallholder farmers. Existing literature widely validates the ISO/IEC 25010 product quality model assessing Functional Suitability, Performance Efficiency, Usability, Reliability, and Maintainability alongside the quantitative, formula-based measurement functions defined in ISO/IEC 25023 for field-deployed embedded systems. Evaluated agricultural IoT applications consistently achieve "strongly acceptable" to "excellent" functionality ratings, deliver high System Usability Scale (SUS) scores among farmers with limited digital literacy, and prove up to 94.21% more cost-effective than traditional labor-based monitoring. Collectively, these reviewed studies validate the architectural foundation of S.A.L.I.M.A.N.G.O., confirming that integrating calibrated low-cost sensors, localized computer vision, autonomous actuation, and solar-powered LoRaWAN telemetry within a structured CDIO framework provides a robust, socio-technically sound solution for mitigating chronic mortality in mud crab farming.

CHAPTER 3:

Research Design

Flowchart of Research Design/Process Flowchart

Description of Research Instrument Used

Controlled Experimental Validation Data Log

To evaluate the technical accuracy and efficiency of the prototype, the researchers will utilize a Controlled Experimental Validation Data Log. This tool is a structured observation sheet used during live, controlled testing of the system at the beneficiary's mud crab farm site. It systematically records the raw, automated telemetry outputs from the multiparameter sensor array (dissolved oxygen, pH, temperature, ammonia, and salinity), the vision system, the actuator control unit, and the LoRaWAN communication module. Specifically, the log is designed to capture paired sensor readings taken simultaneously from the prototype and a calibrated reference instrument for computing percentage accuracy and relative error, biological monitoring events from the vision system categorized as True Positive, False Positive, or False Negative for populating a confusion matrix, timestamped records of hazardous parameter detection versus actuator trigger events for computing detection-to-actuation latency, and packet transmission logs from the LoRaWAN gateway for computing packet delivery rate and average transmission latency. The quantitative data gathered through this instrument directly supplies the variables needed to evaluate the system's performance against the accuracy and efficiency benchmarks established in the

study, and to compare these results with the accuracy levels reported in the related literature reviewed in Chapter 2.

ISO/IEC 25010 Product Quality Evaluation Questionnaire

To determine the level of acceptability of the S.A.L.I.M.A.N.G.O. system, the researchers adapted a structured evaluation survey based on the ISO/IEC 25010 product quality standard, operationalized through the measurement functions defined in ISO/IEC 25023. This questionnaire translates the theoretical quality characteristics of the prototype into concrete, measurable data and is divided into five sections representing the study's target quality traits: Functional Suitability, Performance Efficiency, Usability, Reliability, and Maintainability.

The instrument utilizes a standardized 5-point Likert scale, allowing the beneficiary and other respondents to numerically rate their assessment of the system's hardware, software, and data quality — including the effectiveness of the automated aeration response, the accuracy of the multiparameter sensor readings as presented on the farmer dashboard, and the overall usability of the remote monitoring interface. The numerical weights assigned by the evaluators (ranging from 1 to 5) directly supply the variables x and f required to compute the weighted mean xw, ensuring a mathematically sound and objective interpretation of the prototype's acceptability.

Statistical Treatment

To address the specific research questions outlined in the problem statement and measure stakeholder acceptance of the system, this study will utilize formulas given in the documentation of the sensors present in the study.

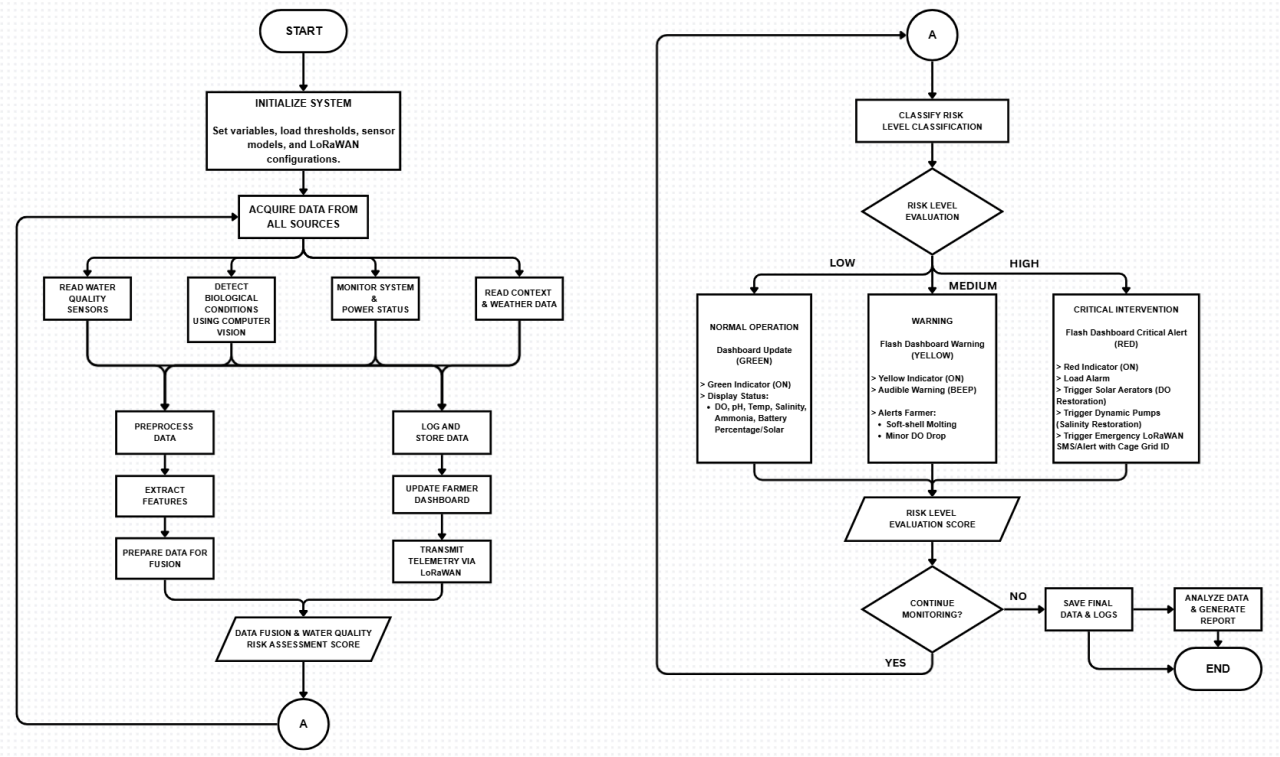
To determine user and stakeholder acceptance of the system's functionality, reliability, and usability, an assessment tool based on a 5-point Likert scale will be utilized. The data collected from these evaluations will then be analyzed by calculating the weighted mean, using the following formula:

$$\bar{x}_w = \frac{\Sigma f x}{N}$$

Where:

- $\bar{x}_w$ = The weighted mean.
- f = The number of respondents for each rating.
- x = The numerical weight assigned to the specific option (1 to 5)
- N = The total number of respondents

Design Project Flow



References

Ahmed, M. A., Gallardo, J. L., Zuniga, M. D., Pedraza, M. A., Carvajal, G., Jara, N., & Carvajal, R. (2022). LoRa based IoT platform for remote monitoring of large-scale agriculture farms in Chile. Sensors, 22(8), 2824.

https://doi.org/10.3390/s22082824

AKVA Group. (2023). AI-driven feeding system. (Company product release). Retrieved from https://www.akvagroup.com

Apine, E., Ramappa, P., Bhatta, R., Turner, L. M., & Rodwell, L. D. (2023). Challenges and opportunities in achieving sustainable mud crab aquaculture in tropical coastal regions. Ocean & Coastal Management, 242, 106711. https://www.sciencedirect.com/science/article/pii/S0964569123002363?via%3Dihub

Bakar, A. A. A., Bakar, Z. A., Yusof, Z. M., Ibrahim, M. J. M., Mokhtar, N. A., & Zaiton, S. N. (2025). IoT-based real-time water quality monitoring and sensor calibration for enhanced accuracy and reliability. International Journal of Interactive Mobile Technologies, 19(1), 155–170.

https://doi.org/10.3991/ijim.v19i01.51101

Global Growth Insights. (2025). IoT for fisheries and aquaculture market size, forecast to 2026. Retrieved from https://www.globalgrowthinsights.com/market-reports/iot-for-fisheries-and-aquaculture-market-100128

Jayasinghe, U., Senarathna, S., & Jayasuriya, D. (2025). Smart sensing and anomaly detection for microalgae culture based on LoRaWAN sensors and LSTM autoencoder. Aquaculture International, 33, 2185. https://doi.org/10.1007/s10499-025-02104-7

Nivas, R., Subramaniam, S., Ponnusamy, V., Ganesan, R., & Kumar, K. S. (2023). Cannibalism and disease transmission in mud crab culture: A critical review. Aquaculture Research, 54(7), 3421-3435.

Olanubi, O. O., Akano, T. T., & Asaolu, O. S. (2024). Design and development of an IoT-based intelligent water quality management system for aquaculture. Journal of Electrical Systems and Information Technology, 11, 12. https://link.springer.com/article/10.1186/s43067-024-00139-z

Roy, D., Padhiary, M., Roy, P., & Barbhuiya, J. (2024). Artificial intelligence-driven smart aquaculture: Revolutionizing sustainability through automation and machine learning. Latia, 2024. Retrieved from https://arxiv.org/pdf/2507.11974

Semtech. (2025). Smart agriculture revolution: How LoRaWAN is transforming farming operations. Retrieved from https://blog.semtech.com/smart-agriculture-revolution-part-1-how-lorawan-is-transforming-farming-operations

Shete, R. P., Bongale, A. M., & Dharrao, D. (2025). IoT-enabled effective real-time water quality monitoring method for aquaculture. Methods X, 13, 102906. https://www.sciencedirect.com/science/article/pii/S2215016124003583?via%3Dihub

Bose, R., Sutradhar, S., Mondal, H., Bhattacharyya, D., & Roy, S. (2024). Integrating environmental monitoring and bird attack prevention in fish farming: A

combined solution for improved pond management. Discover Applied Sciences, 6(3), 81. https://link.springer.com/article/10.1007/s42452-024-05621-x

Chen, X., Zhang, Y., Li, D., & Duan, Q. (2023). Chinese mitten crab detection and gender classification method based on GMNet-YOLOv4. Computers and Electronics in Agriculture, 214, 108318. https://www.sciencedirect.com/science/article/abs/pii/S0168169923007068?via%3Dihub

Eneh, A. H., Udanor, C. N., Ossai, N. I., Aneke, S. O., Ugwoke, P. O., Obayi, A. A., Ugwuishiwu, C. H., & Okereke, G. E. (2023). Towards an improved internet of things sensors data quality for a smart aquaponics system yield prediction. MethodsX, 11, 102436. https://www.sciencedirect.com/science/article/pii/S2215016123004326?via%3Dihub

Feng, R., Zhang, G., Yang, S., & Chen, Y. (2023). Visual detection of Portunus survival based on YOLOv5 and RCN multi-parameter fusion. AgriEngineering, 5(2), 740–760. https://www.mdpi.com/2624-7402/5/2/46

Flores-Iwasaki, M., Guadalupe, G. A., Pachas-Caycho, M., Chapa-Gonza, S., Mori-Zabarburú, R. C., & Guerrero-Abad, J. C. (2025). Internet of Things (IoT) sensors for water quality monitoring in aquaculture systems: A systematic review and bibliometric analysis. AgriEngineering, 7(3), 78. https://www.mdpi.com/2624-7402/7/3/78

Jais, N. A. M., Abdullah, A. F., Kassim, M. S. M., Karim, M. M. A., Abdulsalam, M., & Muhadi, N. A. (2024). Improved accuracy in IoT-based water quality monitoring for aquaculture tanks using low-cost sensors: Asian seabass fish farming. Heliyon.

https://www.cell.com/heliyon/fulltext/S2405-8440(24)05638-X?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS240584402405638X%3Fshowall%3Dtrue

Lindholm-Lehto, P. (2023). Water quality monitoring in recirculating aquaculture systems. Aquaculture, Fish and Fisheries, 3(2), 113–131. https://onlinelibrary.wiley.com/doi/10.1002/aff2.102

Saputra, D. E., Rangkuti, A. H., Dwi Putra, S. E., Daru Kusuma, P., Kurniawan, A., & Gabriela, M. (2025). Automated detection of molting crabs using YOLO: Enhancing efficiency in soft-shell crab aquaculture. JOIV: International Journal on Informatics Visualization, 9(5), 2001–2010.

https://joiv.org/index.php/joiv/article/view/3468

Zhang, Z., Liu, F., He, X., Wu, X., Xu, M., & Feng, S. (2024). Soft-shell crab detection model based on YOLOF. Aquaculture International, 32(4), 5269–5298. https://link.springer.com/article/10.1007/s10499-024-01426-2

Zuhaer, A., Khandoker, A., Enayet, N., Partha, P. K. P., & Awal, M. A. (2025). Sustainable aquaculture: An IoT-integrated system for real-time water quality monitoring featuring advanced DO and ammonia sensors. Aquacultural Engineering, 109, 102678. https://www.sciencedirect.com/science/article/pii/S0144860925001670?via%3Dihub

Alsewari, A. A., & Ali, M. A. H. (2024). Development of LoRaWAN-based IoT system for water quality monitoring in rural areas. *Expert Systems with Applications, 242*, 122862.

https://www.sciencedirect.com/science/article/pii/S095741742303364X?via%3Dihub

Bullecer, B. A. S., Jabao, C. J. S., & Yahya, A. (2025). Remote water quality monitoring system using solar-powered and long-range communication. *Mindanao Journal of Science and Technology, 23*(2), 286–310.

https://mjst.ustp.edu.ph/index.php/mjst/article/view/2501

Chapungo, N. J., & Postolache, O. (2025). Experimental evaluation of LoRaWAN connectivity reliability in remote rural areas of Mozambique. *Sensors, 25*(19), 6027. https://www.mdpi.com/1424-8220/25/19/6027

Durand, T. G., & Booysen, M. J. (2025). Performance evaluation of a mesh-topology LoRa network. *Sensors, 25*(5), 1602.

https://www.mdpi.com/1424-8220/25/5/1602

Espena, G. D., Libao, F. J. D., Comedia, V. J. G., De Luna, N. A. P. U., Mojica, A. J. N., & Rivera, M. D. (2023). Enhancing water quality control and monitoring in shrimp farms with LoRaWAN technology. *2023 IEEE 15th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, 1–6.

https://ieeexplore.ieee.org/document/10589129/

Ijaz, W., & Iqbal, M. T. (2022). Solar powered water pumping system automation and control using a microcontroller for aquaculture. *European Journal of Electrical Engineering and Computer Science, 6*(4), 17–22. https://doi.org/10.24018/ejece.2022.6.4.447

Meza-Solis, F., Cespedes-Ramos, J., Sotomayor-Beltran, C., & Briones-Zuñiga, J. (2025). Automated water quality control system for aquaculture. *International Journal of Engineering Trends and Technology, 73*(2), 200–213.

https://ijettjournal.org/archive/ijett-v73i2p117

Putri, W. M., Syahrir, Nurhanafi, K., Putri, D. R. P. S., Zarkasi, A., & Ummah, A. R. (2026). Arduino-based automatic water salinity control system using a Gravity TDS sensor. *Lontar Physics Today, 5*(1), 115–133.

https://journal3.upgris.ac.id/index.php/lpt/article/view/352

Sangkaphet, P., Chansiri, N., Kaewklom, C., Chaleamwong, B., Wonglamai, P., Chinnachot, P., & Makdee, S. (2026). The development and field evaluation of an IoT–LoRa-based water-quality-monitoring and aeration-actuation system for tilapia cage farming. *Applied Sciences, 16*(11), 5308.

https://www.mdpi.com/2076-3417/16/11/5308

Suriasni, P. A., Faizal, F., Hermawan, W., Subhan, U., Panatarani, C., & Joni, I. M. (2024). IoT water quality monitoring and control system in moving bed biofilm reactor to reduce total ammonia nitrogen. *Sensors, 24*(2), 494.

https://www.mdpi.com/1424-8220/24/2/494à

Adhi, N. S., & Irianto, K. D. (2026). Evaluation of smart agriculture prototype using SUS method. International Journal of Informatics and Computation (IJICOM), 8(1).

https://ijicom.respati.ac.id/index.php/ijicom/article/view/173

Al-Mutairi, A. W., & Al-Aubidy, K. M. (2023). IoT-based smart monitoring and management system for fish farming. Bulletin of Electrical Engineering and Informatics, 12(3), 1435–1446. https://www.beei.org/index.php/EEI/article/view/3365

Analysis of IoT-LoRa to improve LoRa performance for vaname shrimp farming monitoring system. (2024). Jurnal Ilmiah Teknik Elektro Komputer dan Informatika (JITEKI), 10(1). https://journal.uad.ac.id/index.php/JITEKI/article/view/27598

Binayao, R. P., Mantua, P. V. L., Namocatcat, H. R. M. P., Seroy, J. K. K. B., Sudaria, P. R. A. B., Gumonan, K. M. V. C., & Orozco, S. M. M. (2024). Smart water irrigation for rice farming through the Internet of Things. International Journal of Computing Sciences Research, 8, 2550–2563. https://arxiv.org/abs/2402.07917

Erawati, P., Prasti, D., & Kriswinarso, T. B. (2025). Internet of Things (IoT)-based water quality monitoring system design for tilapia fish farming ponds. Brilliance: Research of Artificial Intelligence, 5(2), 860–869. https://jurnal.itscience.org/index.php/brilliance/article/view/6927

Javier, B. S., Paliuanan, L. P., Agpalza, J. K. A., & Agoto, J. S. (2022). MangngalApp – An integrated package of technology for COVID-19 response and rural development: Acceptability and usability using TAM. Journal of Biodiversity

and Environmental Sciences, 21(4), 109–117. https://arxiv.org/abs/2301.02893

Rofiansyah, W., Zalianty, F. R., La Ito, F. A., Wijayanto, I., Ryanu, H. H., & Irawati, I. D. (2025). IoT-based control and monitoring system for hydroponic plant growth using image processing and mobile applications. PeerJ Computer Science, 11, e2763.

https://peerj.com/articles/cs-2763/

Sapin, S. B., Alibudbud, B. A., Molleno, P. B., Veluz, M. B., & Asor, J. R. (2022). Intelligent aquaculture system for pisciculture simulation using deep learning algorithm. Indonesian Journal of Electrical Engineering and Computer Science, 25(1), 561–568. https://ijeecs.iaescore.com/index.php/IJEECS/article/view/26613

Sung, W.-T., Isa, I. G. T., & Hsiao, S.-J. (2023). An IoT-based aquaculture monitoring system using Firebase. Computers, Materials & Continua, 76(2), 2179–2200. https://www.techscience.com/cmc/v76n2/54025

Syed Taha, S. N., Abu Talip, M. S., Mohamad, M., Azizul Hasan, Z. H., & Tengku Mohmed Noor Izam, T. F. (2024). Evaluation of LoRa network performance for water quality monitoring systems. Applied Sciences, 14(16), 7136. https://www.mdpi.com/2076-3417/14/16/7136

Correa, E. B. E., Sousa, J. C., Abelém, A. J. G., & Oliveira, S. R. B. (2022). An evaluation of security features based on ISO/IEC 25023 for a distributed autonomic scientific publisher tool on a permissioned blockchain. *JISTEM – Journal of Information Systems and Technology Management, 19*, e202219020.

https://jistem.tecsi.org/index.php/jistem/article/view/3217

Jeon, M., Kim, J., Kim, S., Park, S., Zhang, B., & Smith, A. H. (2025). *WaggleNet: A LoRa and MQTT-based monitoring system for internal and external beehive conditions* (arXiv:2512.07408). arXiv.

https://arxiv.org/abs/2512.07408

Mnyanghwalo, D., Kundaeli, H., Kalinga, E., & Hamisi, N. (2024). Deployment of an IoT based sensor node for faults detection and classification in electrical secondary distribution networks. *Cogent Engineering, 11*(1), Article 2404679.

https://www.tandfonline.com/doi/full/10.1080/23311916.2024.2404679

Moumane, K., Idri, A., El Aouni, F., Laghnimi, J., Chaouni Benabdellah, N., & Hamal, O. (2024). ISO/IEC 25010-based quality evaluation of three mobile applications for reproductive health services in Morocco. *Clinical and Experimental Obstetrics & Gynecology, 51*(4), 88.

https://www.imrpress.com/journal/CEOG/51/4/10.31083/j.ceog5104088

Royer, L., Terray, L., Rubéo-Lisa, M., Sudre, J., Gauthier, P.-J., Claude, A., Giammanco, S., Pecora, E., Principato, P., & Breton, V. (2024). Lessons learnt from monitoring the Etna volcano using an IoT sensor network through a period of intense eruptive activity. *Sensors, 24*(5), 1577.

https://www.mdpi.com/1424-8220/24/5/1577

Safi, H., Jehangiri, A. I., Ahmad, Z., Ala'anzy, M. A., Alramli, O. I., & Algarni, A. (2024). Design and evaluation of a low-power wide-area network (LPWAN)-based emergency response system for individuals with special needs in smart buildings. *Sensors, 24*(11), 3433.

https://www.mdpi.com/1424-8220/24/11/3433

Sigongan, J. B., Sinodlay, H. P., Cuizon, S. X. P., Redondo, J. S., Macapulay, M. G., Bulahan-Undag, C. O., & Gumonan, K. M. V. C. (2023). GULP: Solar-powered smart garbage segregation bins with SMS notification and machine learning image processing. *International Journal of Computing Sciences Research, 7*, 2018–2036. https://arxiv.org/abs/2304.13040

Singh, [initials not resolved in source]. et al. (2024). Reliability on the Internet of Things with designing approach for exploratory analysis. *Frontiers in Computer Science, 6*, Article 1382347.

https://newinera.com/index.php/JournalLaMultiapp/article/view/1626

Taniasari, N., & Shodikin, M. (2024). Security quality measurement based on ISO/IEC 25023 quality model case study: Hospital management information system. *Journal La Multiapp, 5*(6).

https://newinera.com/index.php/JournalLaMultiapp/article/view/1626

Adhi, N. S., & Irianto, K. D. (2026). Evaluation of smart agriculture prototype using SUS method. *International Journal of Informatics and Computation (IJICOM)*, *8*(1). https://doi.org/10.35842/ijicom.v8i1.173

Al-Mutairi, A. W., & Al-Aubidy, K. M. (2023). IoT-based smart monitoring and management system for fish farming. *Bulletin of Electrical Engineering and Informatics*, *12*(3), 1435–1446. https://doi.org/10.11591/eei.v12i3.3365

Bullecer, B. A. S., Jabao, C. J. S., & Yahya, A. (2025). Remote water quality monitoring system using solar-powered and long-range communication. *Mindanao Journal of Science and Technology*, *23*(2), 286–310. https://mjst.ustp.edu.ph/index.php/mjst/article/view/2501

Chapungo, N. J., & Postolache, O. (2025). Experimental evaluation of LoRaWAN connectivity reliability in remote rural areas of Mozambique. *Sensors*, *25*(19), Article 6027. https://doi.org/10.3390/s25196027

García-Ávila, J., Berra-Davalos, L. D., Hernández-Miranda, V., & Pedroza-Gonzalez, J. (2026). Design for manufacturing in the engineering education 5.0 era: An agile framework for project-based vehicle prototyping. *Frontiers in Education*, *11*, Article 1797638. https://doi.org/10.3389/feduc.2026.1797638

Habbal, F., Kolmos, A., Hadgraft, R. G., Holgaard, J. E., & Reda, K. (2024). *Reshaping engineering education: Addressing complex human challenges*. Springer. https://doi.org/10.1007/978-981-99-5873-3

Ivanov, S., Korchenko, O., Litawa, G., Oliinyk, P., & Oliinyk, O. (2026). Prototype of hydrochemical regime monitoring system for fish farms. *Sensors*, *26*(2), Article 497. https://doi.org/10.3390/s26020497

Jabbar, W. A., Ting, T. M., Hamidun, M. F. I., Kamarudin, A. H. C., Wu, W., Sultan, J., Alsewari, A. A., & Ali, M. A. H. (2024). Development of LoRaWAN-based IoT system for

water quality monitoring in rural areas. *Expert Systems with Applications*, *242*, Article 122862. https://doi.org/10.1016/j.eswa.2023.122862

Mnyanghwalo, D., Kundaeli, H., Kalinga, E., & Hamisi, N. (2024). Deployment of an IoT based sensor node for faults detection and classification in electrical secondary distribution networks. *Cogent Engineering*, *11*(1), Article 2404679. https://doi.org/10.1080/23311916.2024.2404679

Putri, W. M., Syahrir, Nurhanafi, K., Putri, D. R. P. S., Zarkasi, A., & Ummah, A. R. (2026). Arduino-based automatic water salinity control system using a Gravity TDS sensor. *Lontar Physics Today*, *5*(1), 115–133. https://doi.org/10.26877/lpt.v5i1.352

Roy, D., Padhiary, M., Roy, P., & Barbhuiya, J. A. (2024). Artificial intelligence-driven smart aquaculture: Revolutionizing sustainability through automation and machine learning. *LatIA*, *2*, Article 116. https://doi.org/10.62486/latia2024116

Sarwar, A., & Iqbal, M. T. (2022). Solar powered water pumping system automation and control using a microcontroller for aquaculture. *European Journal of Electrical Engineering and Computer Science*, *6*(4), 17–22. https://doi.org/10.24018/ejece.2022.6.4.447

Soto-Rios, P. C., Nagabhatla, N., & Acevedo-Juárez, B. (2023). Circulatory pathways in the water and wastewater sector in the Latin American region. *Water*, *15*(6), Article 1092. https://doi.org/10.3390/w15061092

Zuhaer, A., Khandoker, A., Enayet, N., Partha, P. K. P., & Awal, M. A. (2025). Sustainable aquaculture: An IoT-integrated system for real-time water quality monitoring featuring advanced DO and ammonia sensors. *Aquacultural Engineering*, *112*, Article 102620. https://doi.org/10.1016/j.aquaeng.2025.102620